

AZIZA AMAKASSOU

Final-Year Engineering Student in Big Data, Data Science & Artificial Intelligence
Expected Graduation: June 2027

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Final-year engineering student (Bac+5) specializing in **Big Data, Data Science and Artificial Intelligence** at ENSIASD, with a strong interest in data engineering and the design of intelligent algorithmic solutions. Focused on data processing and on optimizing decision-support systems to turn complex problems into performance gains.

EDUCATION

ENSIASD – Taroudant, Morocco 2024 – june 2027
École Nationale Supérieure de l'Intelligence Artificielle et des Sciences des Données
Final year of the engineering degree (Bac+5): Data Science, Big Data and Artificial Intelligence.

EST Essaouira, Morocco 2022 – 2024
École Supérieure de Technologie d'Essaouira, Cadi Ayyad University
University Diploma of Technology (DUT, Bac+2) in Business Intelligence and Data Science.

Moroccan Baccalauréat in Applied Mathematical Sciences 2022
Lycée Sidi Daoud, Ouarzazate.

PROFESSIONAL EXPERIENCE (INTERNSHIPS)

Data Engineer /BI/AI Intern – ORMVAO, Ouarzazate August – September 2026
regional agricultural development authority .
Automated decision-support chain for budget execution monitoring.

- Built an ETL pipeline (Python, Pandas, pyodbc) consolidating heterogeneous Excel sources into a SQL Server data warehouse.
- Delivered a Streamlit application that generates the 8 statutory budget reports in Excel (openpyxl, live formulas, official template), with Plotly dashboards for single- and multi-year budget analysis, shipped as a standalone package for non-technical staff.
- Integrated a conversational AI assistant answering budget execution questions with figures computed directly from the loaded data, so its answers always match the official reports.

AI / Web Development Intern – ENSIASD, Taroudant June – July 2026
Web application for AI-based timetable generation and management.

- Developed a FastAPI / PostgreSQL / React-TypeScript platform to manage teachers, rooms, programs and groups, with Excel import and conflict detection.
- Designed a constraint-based optimization engine with Google OR-Tools (CP-SAT), compared with metaheuristics (tabu search, simulated annealing).

Computer Vision / Deep Learning Intern – AEM, Aït Melloul 2025
Aït Melloul Électromécanique, Moroccan manufacturer of agri-food processing machinery. Automatic tomato sorting (healthy / reject) using computer vision.

- Trained a CNN (TensorFlow/Keras) on a hybrid dataset (Kaggle + real images from AEM sorting machines): 88% overall accuracy.
- Built a Flask web app that detects tomatoes, classifies each fruit and extracts its average hue.

Data / BI Intern – CCIS, Essaouira 2024
Chambre de Commerce, d'Industrie et de Services (Essaouira branch). Customer management application to automate file tracking.

- Developed a Java desktop application (JavaFX, JDBC, MySQL): authentication, customer and company forms, printing and Excel export (Apache POI).

Web Development / Data Engineering Intern – AFRIKA ENTOURAGE, Agadir 2023
IT and security equipment company. Web application for farm expense management and tracking.

- Developed a Django web application (Python, HTML/CSS) to track products, purchases, stock outflows and suppliers, with search by period.

ACADEMIC PROJECTS

Intelligent Big Data Platform for Diabetes Risk Analysis and Prediction 2026
Big Data Module Project – ENSIASD

- Designed and developed an end-to-end Big Data pipeline for healthcare analytics focusing on diabetic patients, processing over 25,000 synthetic clinical records.
- Implemented data ingestion, transformation, and validation pipelines using **Logstash**, handling error detection and anomaly management.
- Configured explicit mappings and indexed structured healthcare data into **Elasticsearch** for optimized querying and analytics.
- Built operational dashboards and performed visual data exploration using **Kibana**.
- Prepared and structured data pipelines for distributed data processing and predictive analytics using **Apache Spark**.

SafeNow – Emergency & Safety Assistant (Mobile Application) 2026
End-of-module project in Android mobile development and immersive systems.

- SOS button that sends the user's GPS position (Google Maps link) by SMS to emergency contacts, even without Internet access; contact management, alert history, Firebase authentication.
- Virtual reality extension built with Unity 3D: evacuation simulation in a 3D environment.

Emotion Detection in Reddit Comments 2025
End-of-module project in object-oriented programming (Java) and natural language processing (NLP).

Morocco's Energy Consumption Prediction 2025 – 2026
End-of-module project in Machine Learning.

- Compared linear regression, Random Forest and XGBoost on consumption and weather data with a chronological 80/20 split: XGBoost selected ($R^2 = 0.98$).
- Built a Streamlit dashboard simulating weather scenarios, with a load-shedding risk alert.

Real-Time American Sign Language (ASL) Recognition 2024
DUT final-year project.

- Classified signs from webcam images: hand detection (OpenCV, cvzone), model trained on 300 images per class, Flask web app for real-time prediction.

TECHNICAL SKILLS

Data Science & AI: Machine Learning, Deep Learning, Data Mining, NLP, LLM, RAG, embeddings, AI Agents

Big Data & Databases: Spark, Hadoop, Elasticsearch, Logstash, Kibana, MongoDB, Oracle, SQL Server, PostgreSQL, MySQL, SQL

Programming & Development: Python, PySpark, Java, C, PHP, Flask, Django, React, HTML/CSS, Android

Data Engineering & Tools: ETL, Data Warehouse, Power BI, Streamlit, Android Studio, Google OR-Tools (CP-SAT), Unity 3D, Git, Linux

Modeling: UML, Merise

LANGUAGES

Arabic: Native **French:** Fluent **English:** Fluent